Akina Kuperus

PhD Candidate, Mathematical Biology
University of Victoria
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EDUCATION

PhD Candidate in Mathematical Biology - University of Victoria, Victoria BC
September 2024 - Present
Supervisors: Dr. Mark Lewis & Dr. John Fryxell

MSc in Mathematics - University of Victoria, Victoria BC
September 2021 - August 2023
Supervisor: Dr. Natasha Morrison
Thesis: Cross-Sperner Systems

BSc Honours in Mathematics - University of Regina, Regina SK
September 2017 - May 2021

GIS Data and Technologies Micro-Certificate - Seneca Polytechnic, Toronto ON
December 2021 - February 2022
Demonstrated the ability to use data collection technologies such as ESRI's Field Maps, Survey123, drone-based data collection, etc., as it relates to GIS.

Math to Power Industry - Pacific Institute of Mathematical Science
July 2024 - August 2024
This project paired our team of mathematicians with an industry partner to develop an algorithm to determine construction requirements for a building based on the BC building code. This project relied on mathematical, programming, and professional skills.

PUBLICATIONS

Natalie Behague, **Akina Kuperus**, Natasha Morrison, and Ashna Wright. Improved bounds for cross-Sperner systems. Electron. J. Combin.31(2024), no.2, Paper No. 2.4, 14 pp.

WORKSHOPS, SUMMER SCHOOLS & MEETINGS

Mathematics of Human-Environmental Systems and Climate Change Impacts, University of Victoria
August 2026
This summer school focused on the mathematics of human-environmental systems and climate change impacts. The program involved lectures by Jimmy Garnier, Frank Hilker, Péter K Molnár, and Jody Reimer, as well as a group project applying Markov decision processes to fisheries management.

MERGE workshop, University Center in Svalbard
March 2024
This workshop merged ecological and evolutionary methods to understand adaptation to climate change in Svalbard's tundra ecosystem by bringing together experts in ecology, evolution, genomics, and statistics.

Maud Menten Institute, Pacific Institute for the Mathematical Science Research Network
September 2024 - Present
Member of the collaborative research platform for mathematical biologists at PIMS sites to promote interactions with life science experts and decision makers in government, industry and NGOs.

Mathematical Science Research Institute Random Graph Summer School, Berkeley, California
July 2022
Two week lecture series taught by Prof. Dr. Remco William van der Hofstad on random graph models and their properties.

SCHOLARSHIPS & AWARDS

Biology Graduate Symposium: Best 12-min PhD talk - April 2026

Jarmila Vlasta Von Drak Thouvenelle Graduate Scholarships - 2025
Valued at $5,000

NSERC Canada Graduate Scholarship - Doctoral - 2025 to 2028
Valued at $120,000

University of Victoria Graduate Fellowship (Doctoral) - 2025
Valued at $4,500

University of Victoria Fellowship - Doctoral - 2024, 2025
Valued at $2,000

David F. Strong Research Scholarship - 2022
Valued at $1,400

Canada Graduate Scholarships Master's NSERC - 2022
Valued at $17,500

President's Research Scholarship - 2022, 2025, 2026
Valued at $5,000

University of Victoria Graduate Entrance Award - 2021
Valued at $5,000

University of Victoria Graduate Award - 2021
Valued at $1,500

High Honours in Mathematics from University of Regina - 2021
Awarded to the students who take a more intensive concentration in their major.

Governor General's Academic Silver Medal - 2021
Awarded to the undergraduate student achieving the highest academic standing upon graduation.

University Prize in Science - 2021
Awarded to the most distinguished student at the University of Regina in the Faculty of Science based on academic achievement during the previous twelve months.

Academic Gold Scholarship - 2018, 2019, 2020
Valued at $1,500. Awarded to students with a minimum GPA of 95/100 during the previous academic year.

James B. Carefoot Memorial Scholarship - 2019
Valued at $700.

Joan & Norman Beirnes Scholarship - 2019
Valued at $7,500.

Interpretation Canada Gold Medal - 2019
Award the best personal interpretation program by Interpretation Canada for the Parks Canada program "Game of Homes" in Banff National Park.

PRESENTATIONS

Should I stay or should I go? Modelling mother-calf movement in Svalbard reindeer - April 2026
Biology Graduate Student Symposium, University of Victoria
Presentation of novel coupled step selection function. Winner of best PhD talk.

The application of step selection functions to Svalbard Reindeer - July 2025
Maud Menten Institute, Society for Mathematical Biology, Gordon Research Conference: Movement Ecology
Poster presentation on parameter estimation techniques for step selection functions and description of future work applying step selection functions to mother-calf movement of Svalbard reindeer.

Keeping up with the Cross-Sperners: An Introduction to Cross-Sperner Systems - March 2023
University of Victoria Discrete Math Seminar
Lecture about my research on cross-Sperner systems given as a part of the discrete math seminar series.

An Application of the Hypergraph Container Method - February 2022
University of Victoria
Presentation given at the combinatorics mini-seminar on using the hypergraph container method to count the number of metrics on a graph.

Cross-Sperner Family Pizza Party - December 2022
Pacific Rim Mathematical Association Congress
Poster presentation on my research regarding bounding the size of cross-Sperner families.

Normalized Matching Property in Posets - October 2022
University of Victoria
Presentation given at the combinatorics mini-seminar on the normalized matching, LYM, and chain covering property of partially ordered sets.

Some Applications of Entropy to Combinatorics - June 2022
University of Victoria
Based on the "Three tutorial lectures on entropy and counting" by David Galvin, this presentation was given at the combinatorics mini-seminar highlighting the applications of entropy to combinatorial problems.

A Review and Proof of the Classical Brouwer's Fixed Point Theorem and Schauder's Extension - April 2021
University of Regina
Honours lecture on the famous fixed point theorem for finite dimensional spaces and its extension to infinite dimensional spaces (presented virtually).

On the Universal Surjectivity of the Cantor Set - December 2020
University of Regina
Honours lecture on the Hausdorff-Alexandroff theorem and its application to space-filling curves (presented virtually).

WORK EXPERIENCE

Teaching Assistant, University of Victoria
September 2024 - May 2025
Delivered weekly tutorial sessions for Calculus 1 and 2 students.

Sessional Instructor, University of Victoria
September 2023 - December 2023
Mathematics & Statistics Department
Teaching undergraduate finite math to 100 students. Curriculum included probability, linear programming, and Markov chains.

Interpretation Supervisor, Parks Canada
April 2023 - August 2023
Banff National Park
Supervise interpretation team, write and deliver programs for visitors of the park on ecology and cultural history. Manage multiple projects simultaneously, organize schedules and training for staff, and collaborate with stakeholders.

Math Tutor, University of Victoria
January 2023 - April 2023, September 2024 - May 2025
Math and Stats Assistance Center
Drop in help center for students to receive free tutoring in their 100/200 level math and stats classes.

Interpreter, Parks Canada
Summer 2018, 2019, 2020, 2021
Banff National Park
Deliver programs on ecology and cultural history of Banff to a general audience. Trained in dialogic and inspiration-based interpretation, as well as implementing a growth mindset in the workplace.

Teaching Assistant, University of Regina
September 2019 - April 2020
Linear Algebra, Calculus 1 & 2
Delivered weekly 50 minute tutorial session for 50 to 75 students. Explained practice problems, answered questions, and administered quizzes.

OTHER EXPERIENCE

Math Camp Instructor (Volunteer), University of Regina
September 2020 - December 2020
Delivered hour long extra curricular math sessions to 20 Grade 5/6 students at the university. Designed activities to encourage curiosity in math.